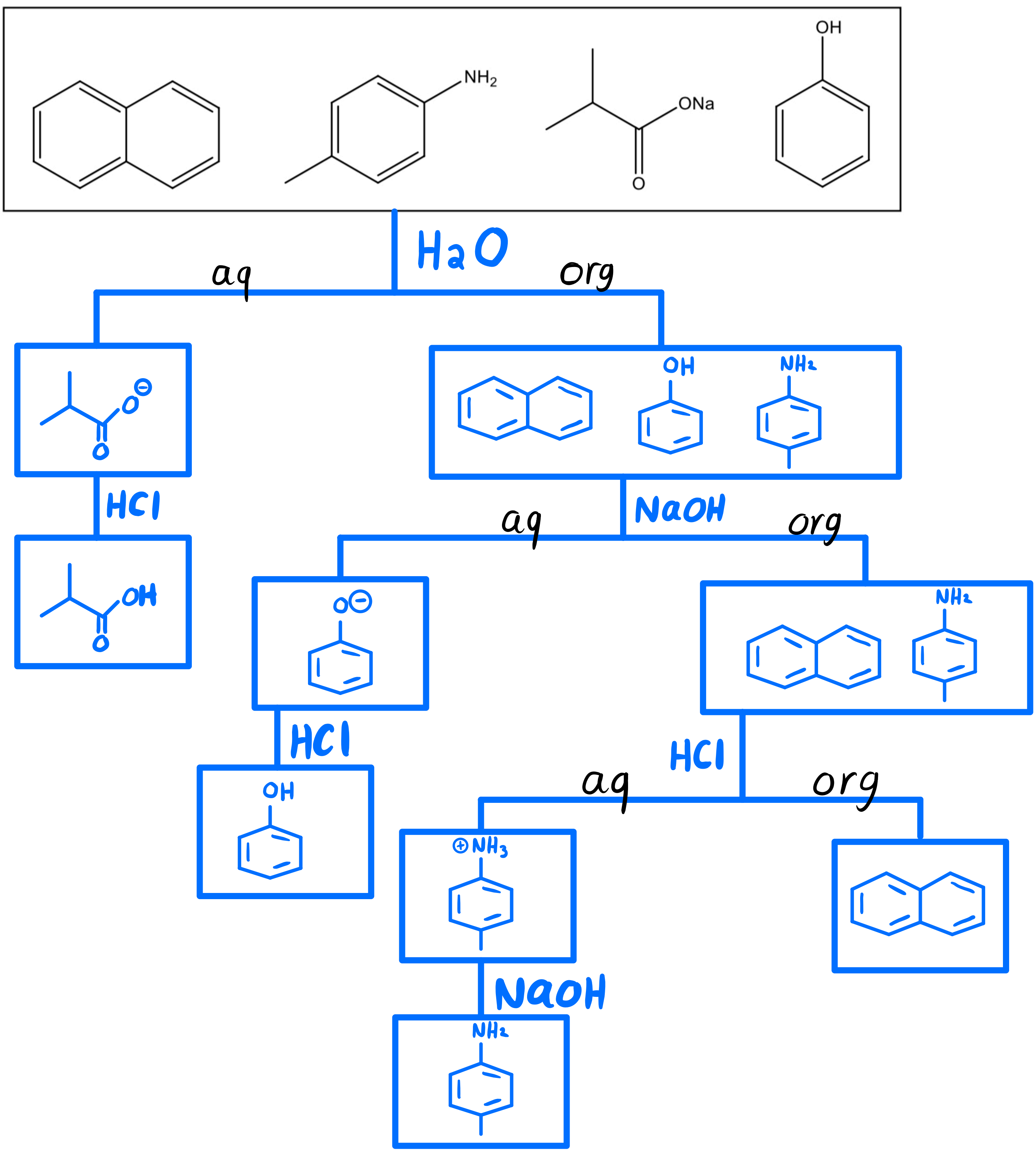
Midterm Review

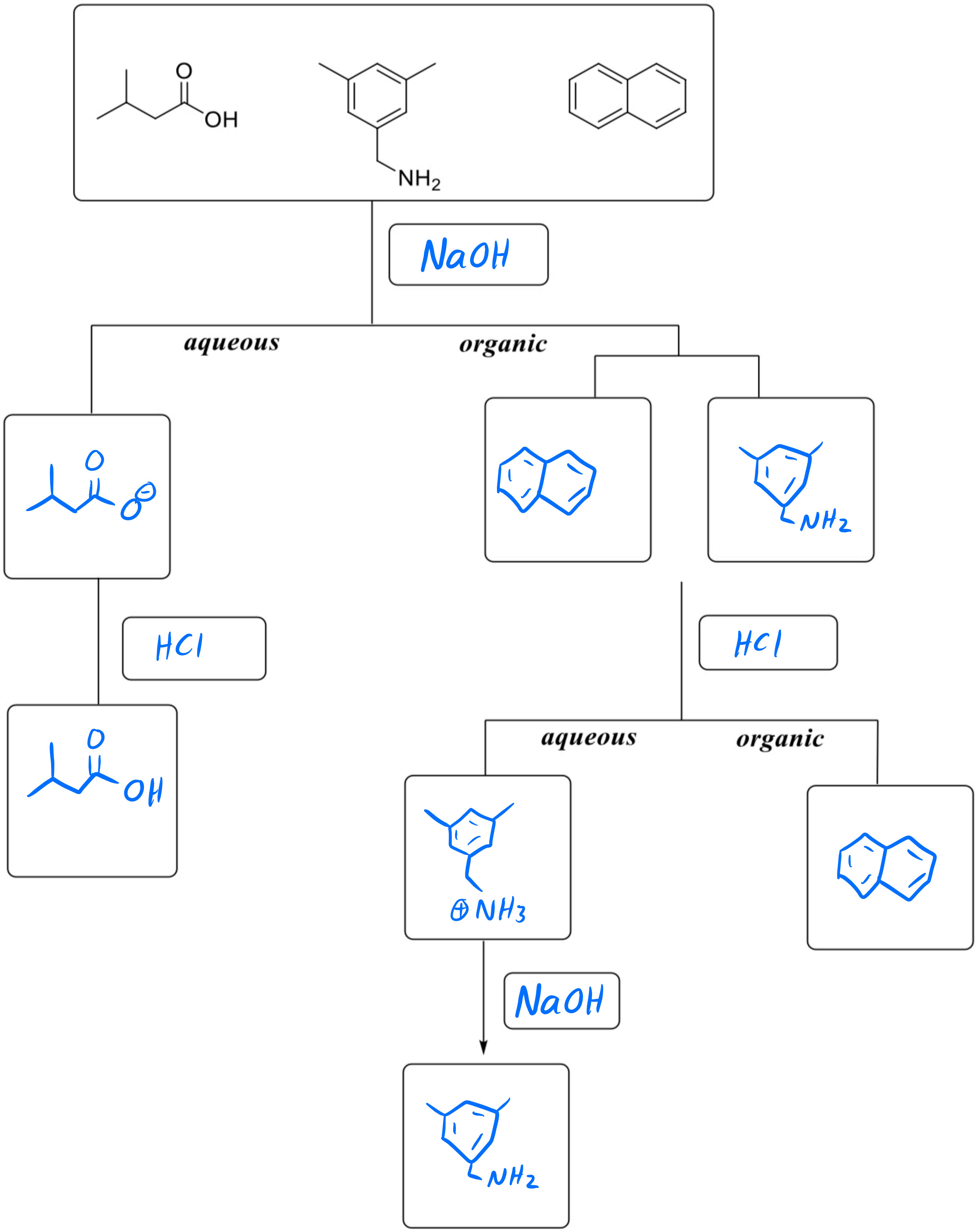
Key

Extraction & Recrystallization

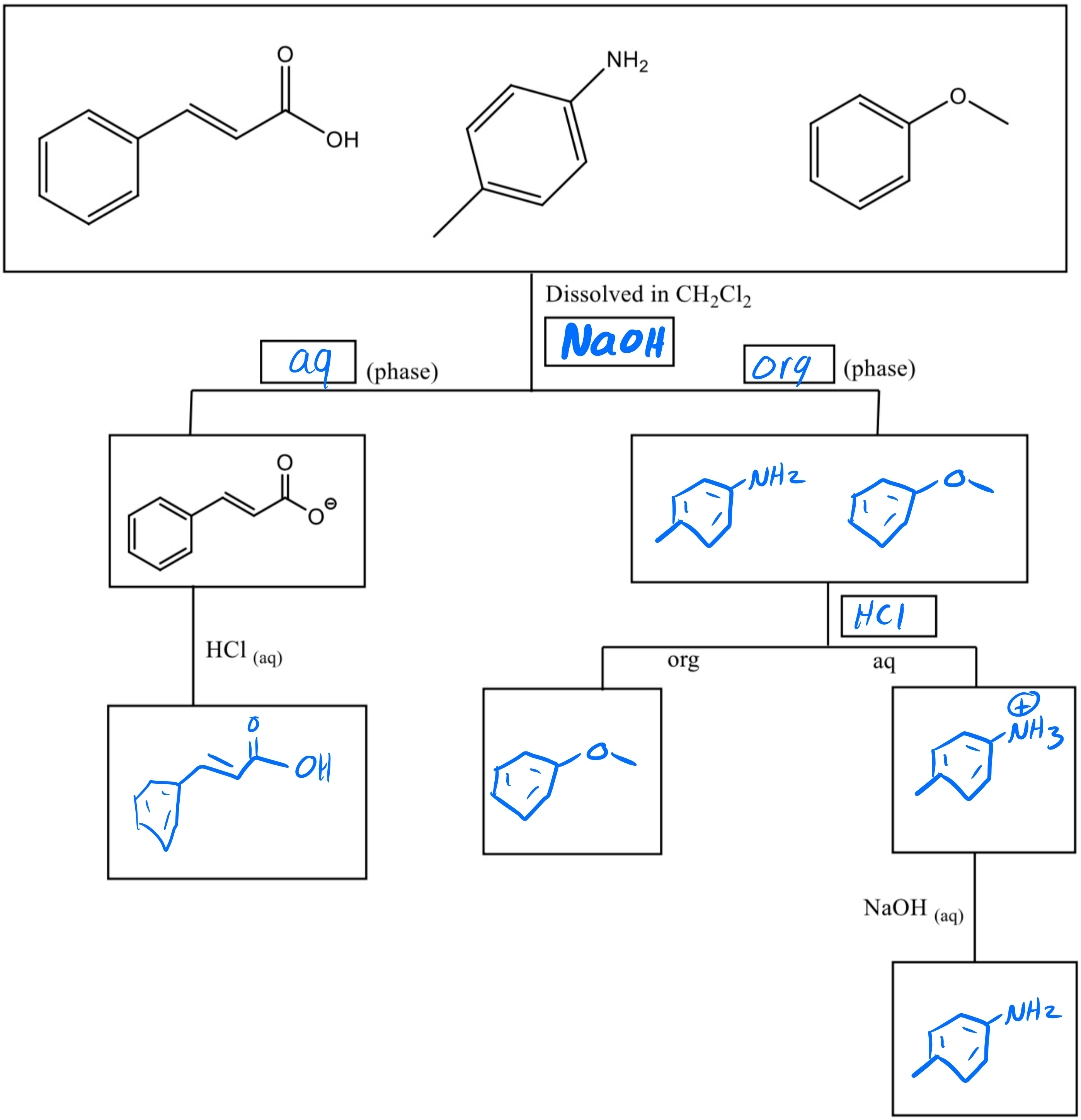
1. Draw an extraction diagram showing how to separate the following mixture. Assume the mixture is dissolved in some organic solvent.



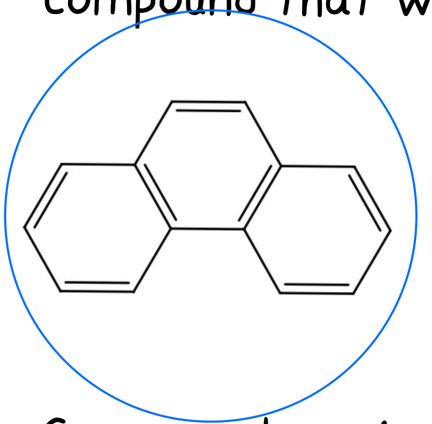
2. The following mixture is dissolved in diethyl ether. Provide an extraction scheme that shows how these three compounds can be separated using only HCl or NaOH. Draw the structures in the boxes.

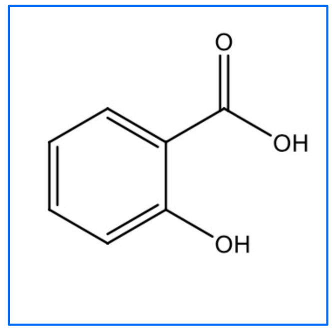


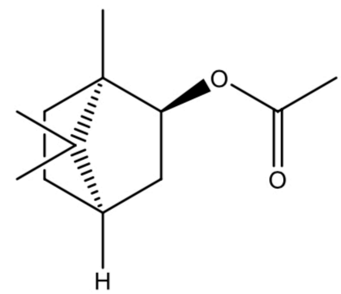
3. The following mixture is dissolved in dichloromethane. Fill in the missing reagents (HCl or NaOH), phases (organic or aqueous), and draw the structures in the boxes.



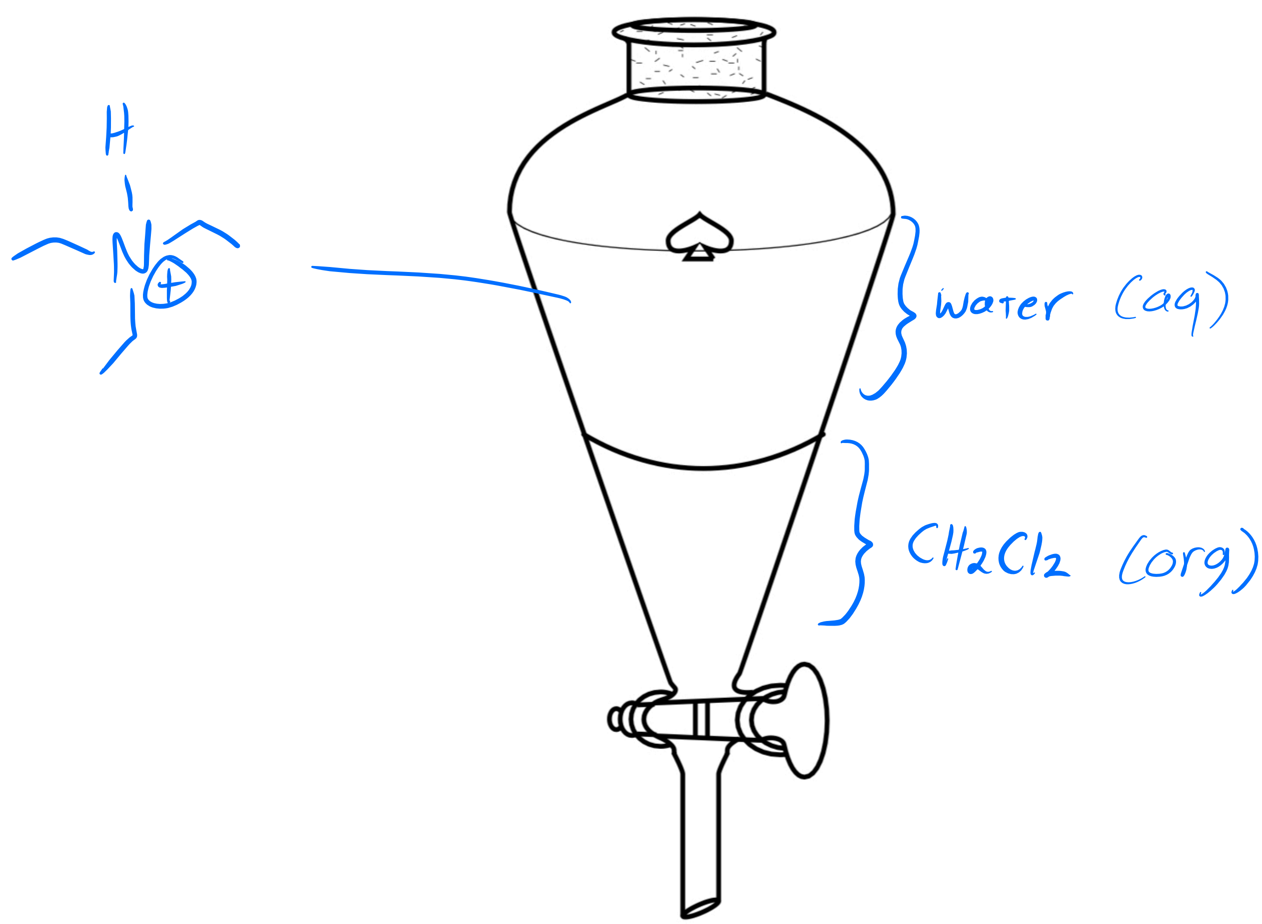
4. Given the $K_D = [X]_{organic}/[X]_{aqueous}$ circle the compound that would exhibit the largest K_D. Draw a square around the compound that would exhibit the smallest K_D.







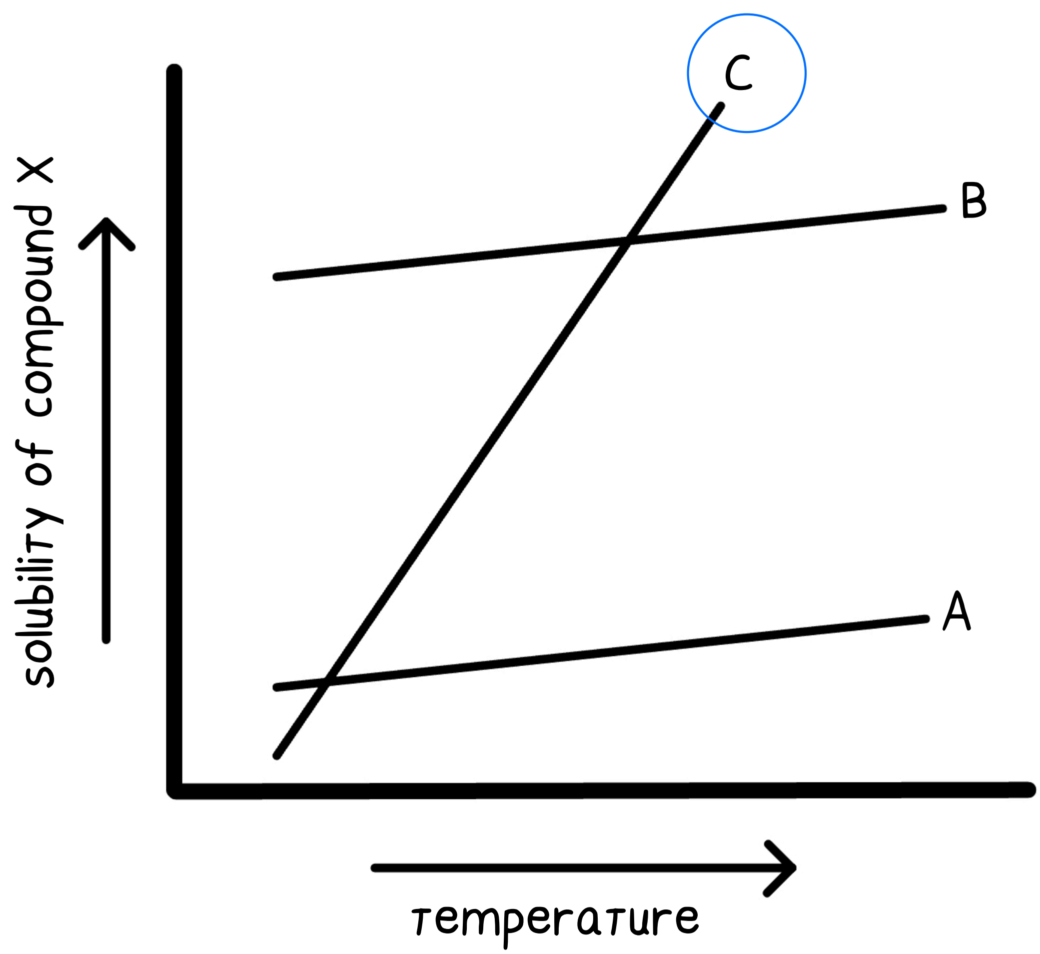
5. Suppose there is a 100 mL solution of triethylamine, $(CH_3CH_2)_3N$, in dichloromethane (CH_2Cl_2). The density of dichloromethane is 1.33 g/mL. To the solution is added 100 mL of 1 M $HCl_{(aq)}$ and the combined volume is transferred to a separatory funnel. Label each layer of the separatory funnel as aqueous or organic. In what layer is the amine?



6. Determine the ideal recrystallization solvent (methanol, hexanes, 1:1 hexanes:methanol) for the purification of compound A given the following solubilities for compound A and impurities B and C.

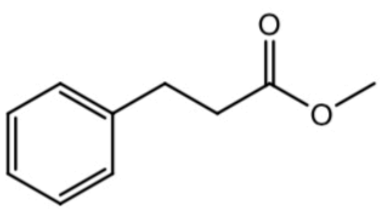
Circle One→	**Methanol**	**Hexanes**	**1:1 Hexanes:methanol**
Room Temp.	**A** – 1 g/L **B** – 1,500 g/L **C** – 0 g/L	**A** – 0.01 g/L **B** – 0 g/L **C** – 600 g/L	**A** – 80 g/L **B** – 20 g/L **C** – 800 g/L
Solvents B.P.	**A** – 100 g/L **B** – 1,600 g/L **C** – 0 g/L	**A** – 0.05 g/L **B** – 0 g/L **C** – 800 g/L	**A** – 85 g/L **B** – 25 g/L **C** – 1,000 g/L

7. Which of the three solvents (A, B, C) would be desirable for the recrystallization of compound X?

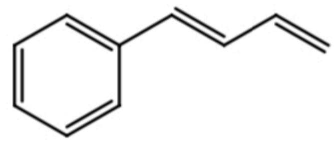


Chromatography

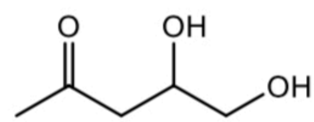
8. The TLC plate below shows three spots after development. Assume the mobile phase movement is from the bottom to the top of the plate. Match the compounds (A, B, C) with their respective spots on the TLC plate based on relative polarity.



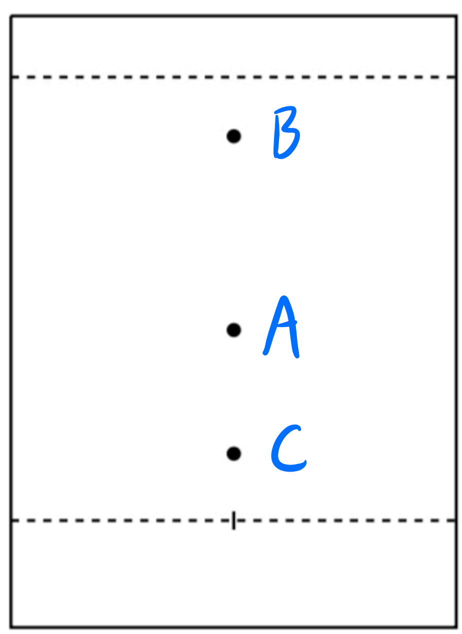
A



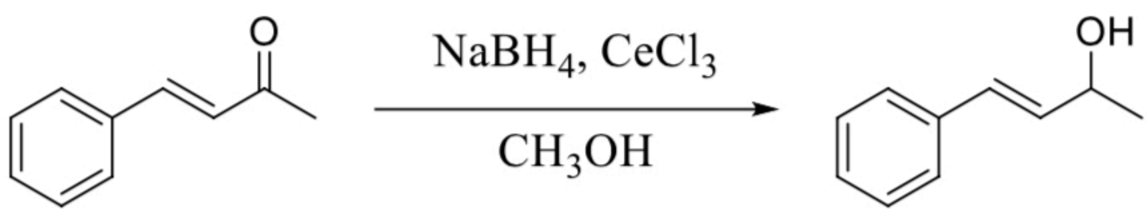
B

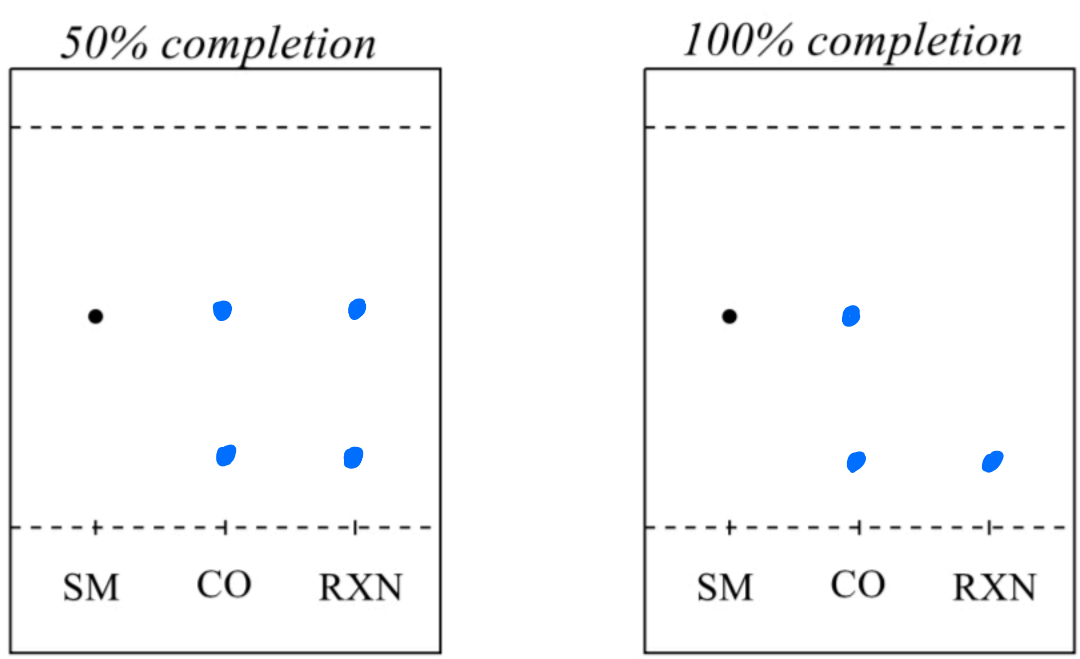


C

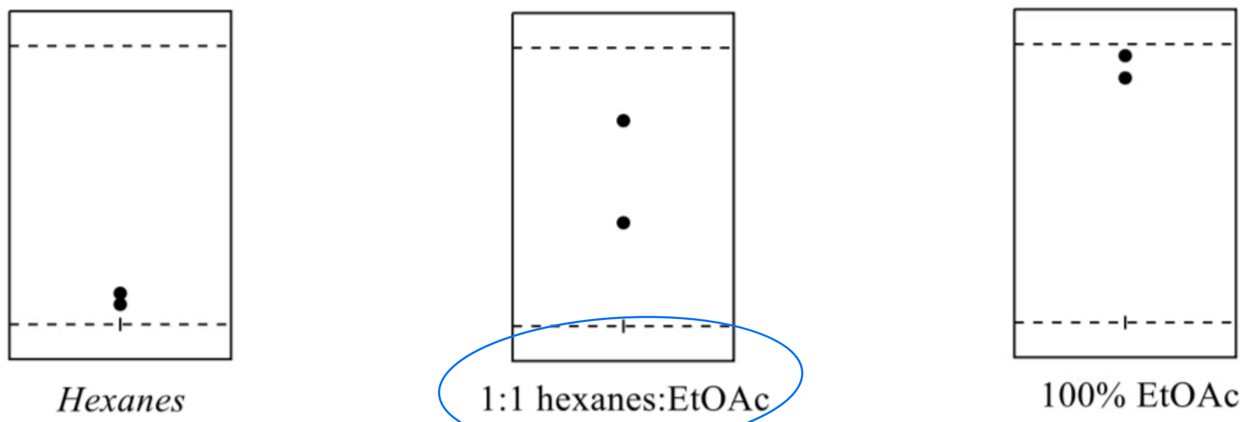


9. For the following reaction, TLC analysis was performed to monitor the progress of the reaction using a silica plate. Visualization of the organic compound spots on a TLC plate can be made upon exposure of each solvent-dried plate to a UV lamp. On each TLC plate, a student has placed a sample of the starting material (SM) as a reference on the left side of the plate, a spot of the reaction mixture (RXN) on the right, and a co-spot (Co) in the center of each. Fill in the spots that would be expected when the reaction is 50% complete and 100% complete.

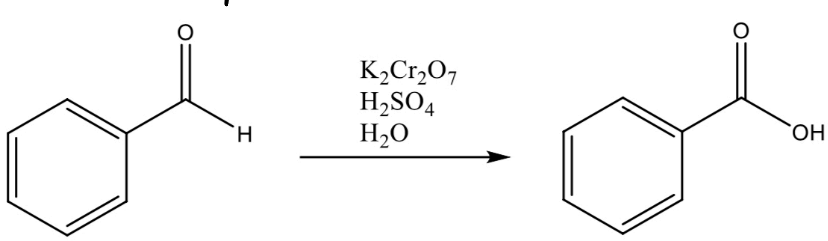


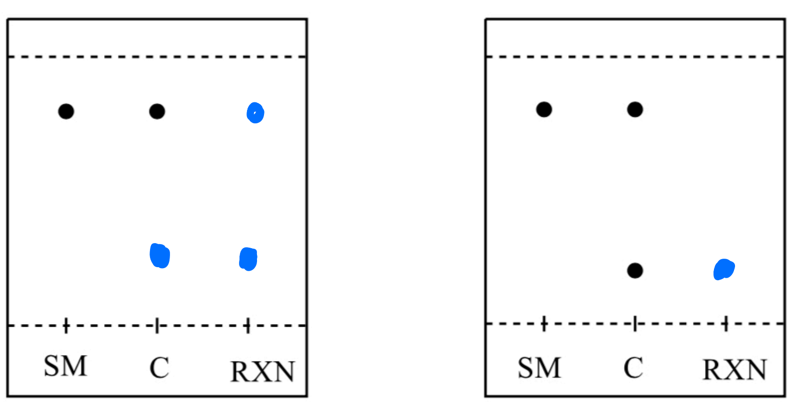


10. A 2-compound mixture was spotted on three separate TLC plates. Each plate was developed in one of three different solvents (hexanes, 1:1 hexanes:Ethyl acetate, and 100% ethyl acetate). Which solvent is the best mobile phase for developing?



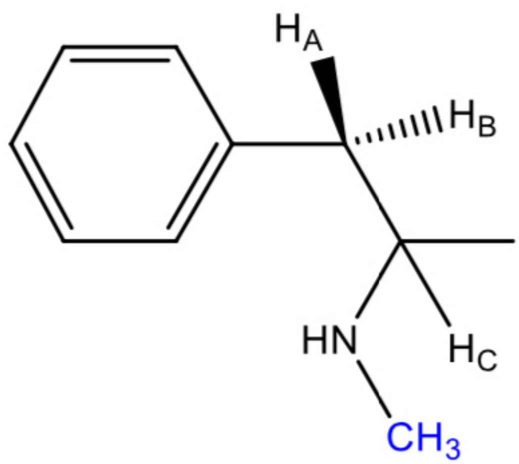
11. The following reaction is being monitored using TLC. The following TLC plates represent 50% completion (left plate) and 100% completion (right plate) of the reaction. Draw in the missing spots to show what the plate would look like at 50% completion and 100% completion. Assume the reagents above the arrow won't show on the TLC plate. SM is starting material, C is a co-spot, and RXN is a sample from the reaction mixture.





Spectroscopy

12. Consider the following molecule:

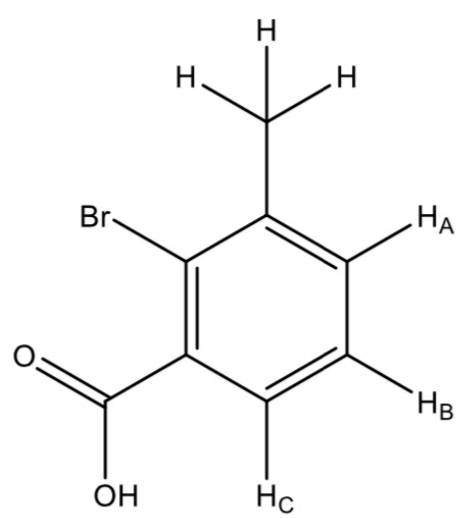


A. What is the splitting pattern of the signal for Ha if $J_{2\text{-bond}}$ is greater than $J_{3\text{-bond}}$? doublet of doublets

B. What is the splitting pattern of the signal for Ha if $J_{2\text{-bond}}$ is equal to $J_{3\text{-bond}}$? triplet

C. What is the expected splitting pattern of the blue methyl protons? singlet

13. Consider the following molecule:

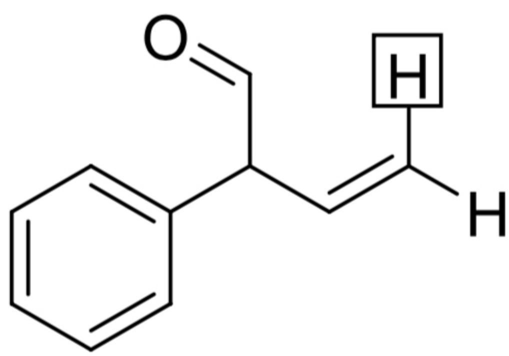


A. What is the splitting pattern of the signal for Hb if J_{BA} is equal to J_{BC}? Triplet

B. What is the splitting pattern of the signal for Hb if J_{BA} is greater than J_{BC}? doublet of doublets

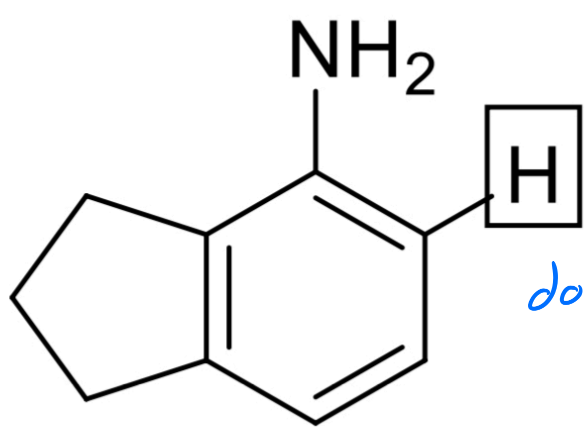
C. What is the splitting pattern of the signal for Ha if $J_{3\text{-bond}}$ is greater than $J_{4\text{-bond}}$? doublet of doublets

14. What is the expected splitting pattern of the indicated proton if $J_{2\text{-bond}}$ is equal to $J_{3\text{-bond}}$?



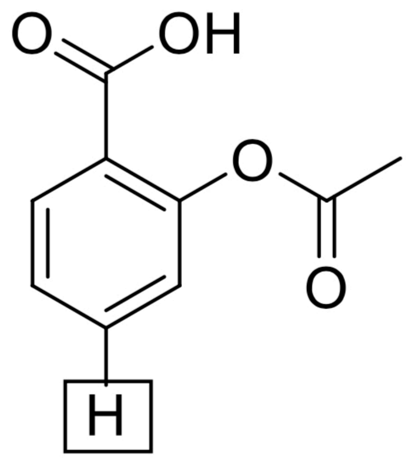
Triplet

15. What is the expected splitting pattern of the indicated proton if 3- and 4-bond coupling is observed?



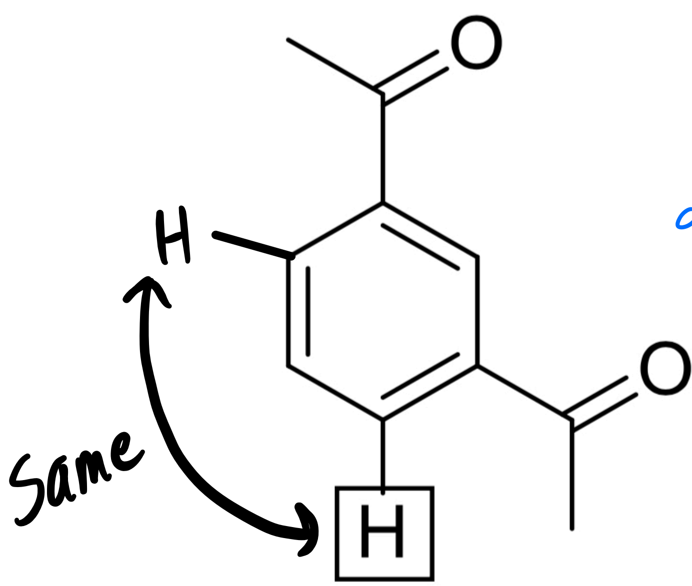
doublet of doublets

15. What is the expected splitting pattern of the indicated proton in the following molecule if all of the coupling interactions have different coupling constants?



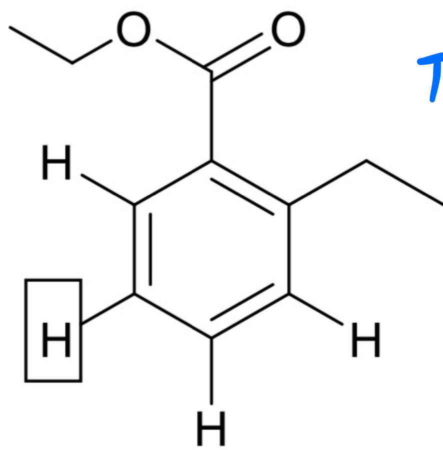
ddd

16. What is the expected splitting pattern of the indicated proton in the following molecule if all of the coupling interactions have different coupling constants?



doublet of doublets

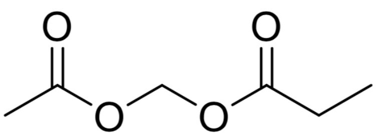
17. What is the expected splitting pattern of the indicated proton in the following molecule if $J_{3\text{-bond}}$ are all equal but larger than $J_{4\text{-bond}}$?



Triplet of doublets

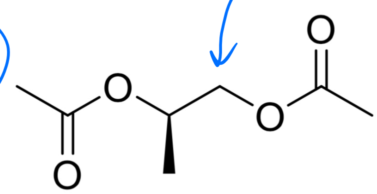
18. How many chemically unique protons are there in each of the following compounds?

4



6

diastereotopic



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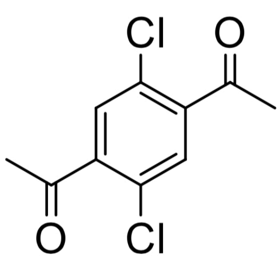

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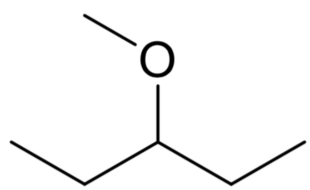

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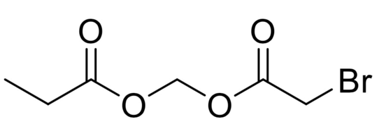

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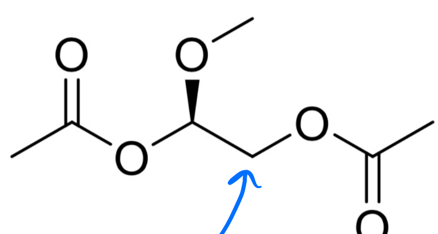
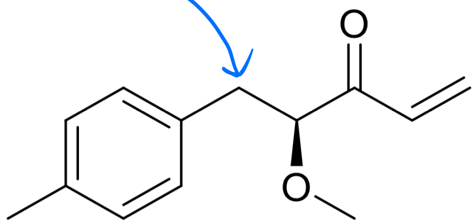


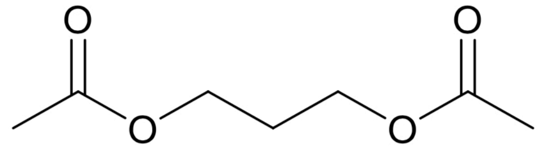
diastereotopic

diastereotopic

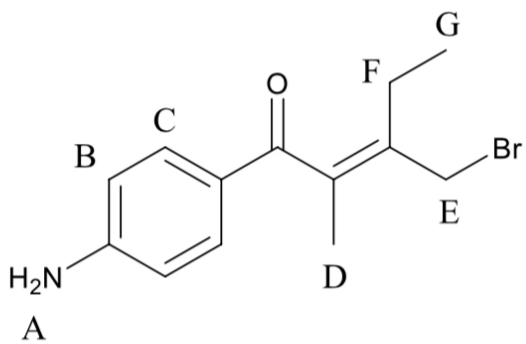
10

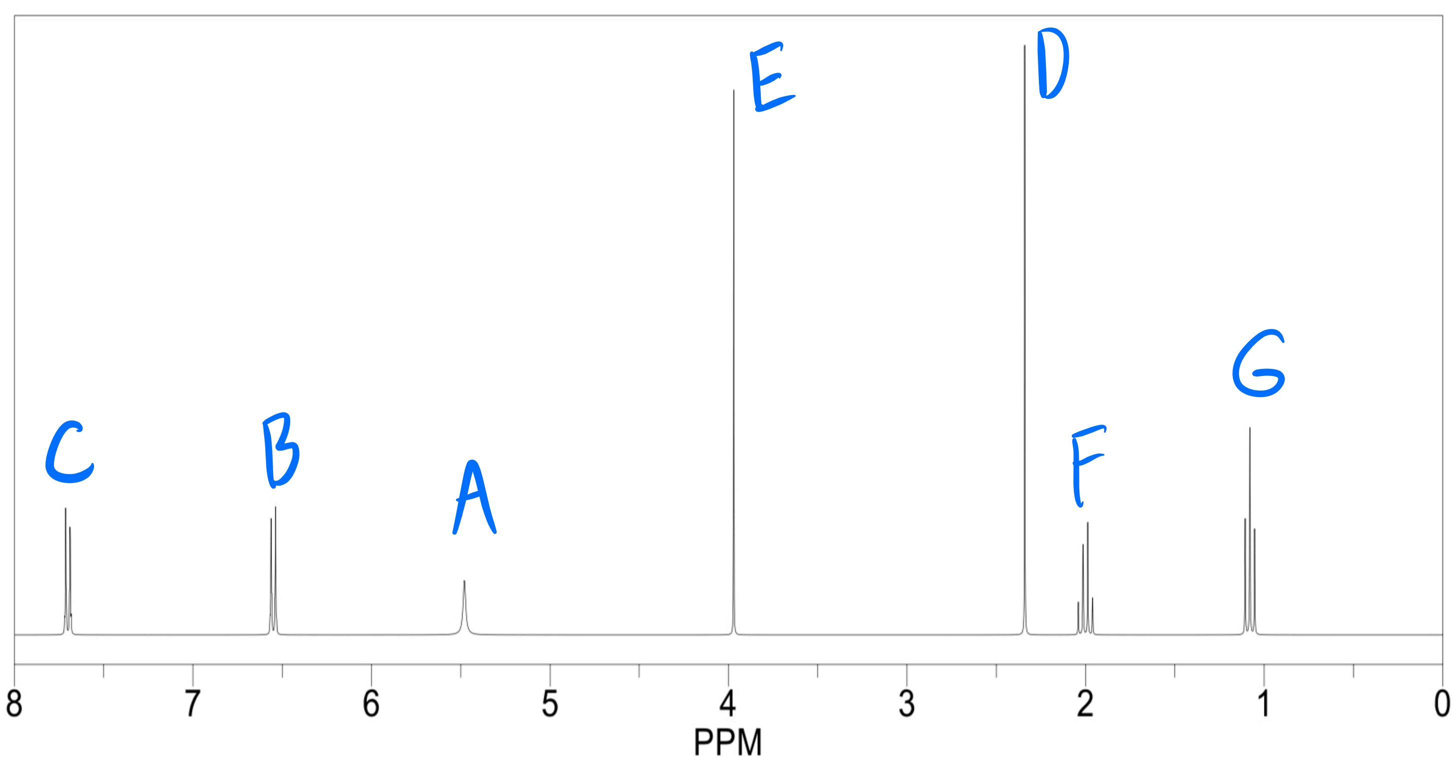


3

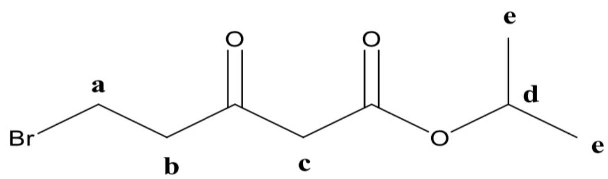


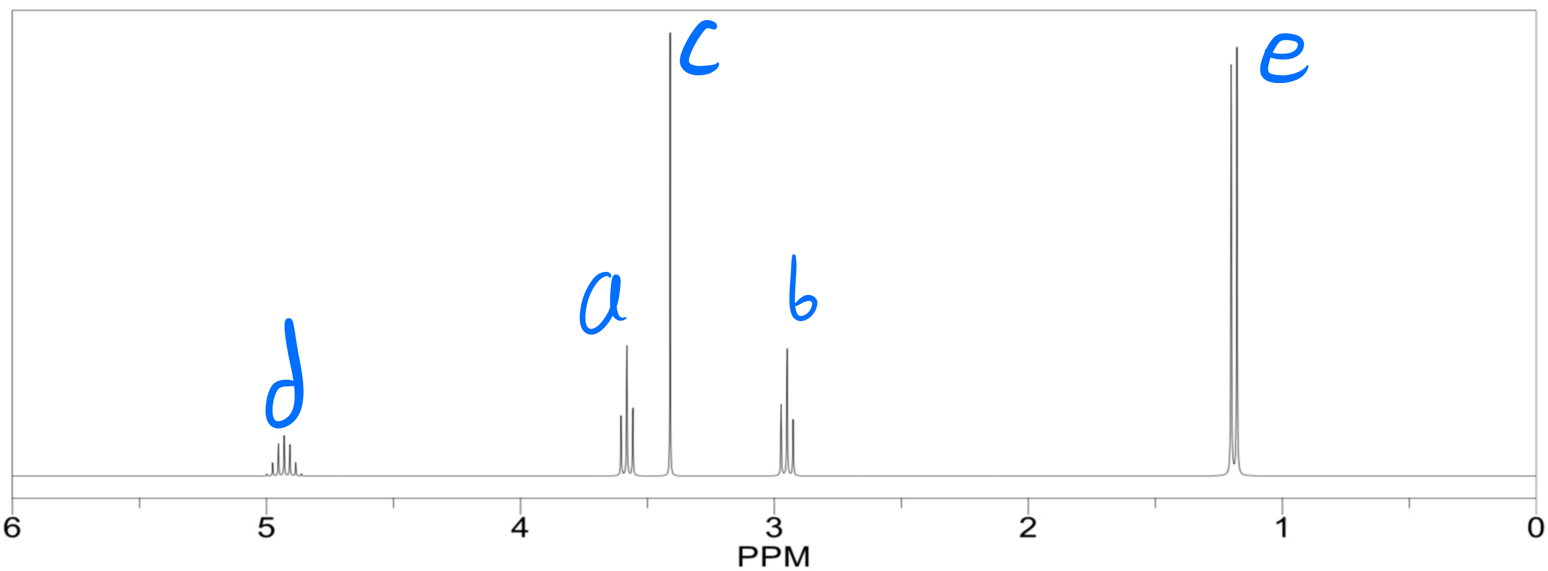
19. Assign the protons (labeled A-G) of the structure below to their corresponding signal on the NMR spectrum. Note: D is not deuterium. That is a methyl group and each proton of the methyl group is labeled D.



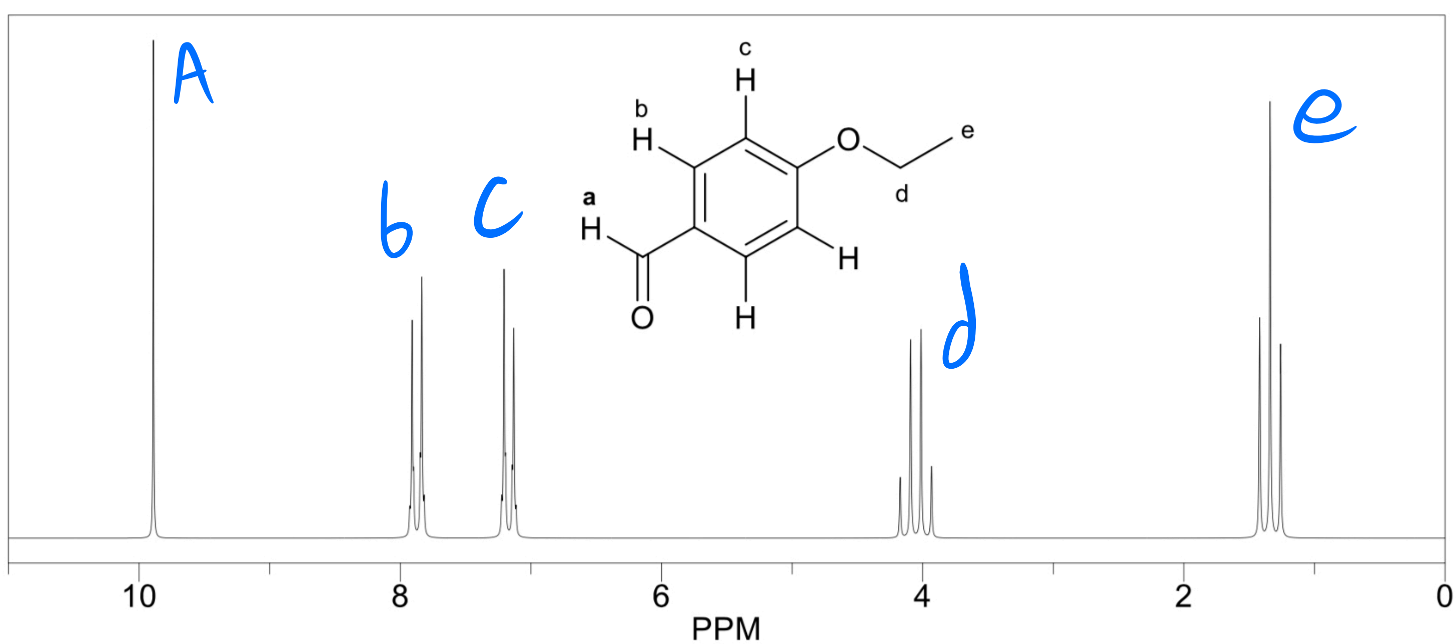


20. Assign the protons (labeled A-E) of the structure below to their corresponding signal on the NMR spectrum.

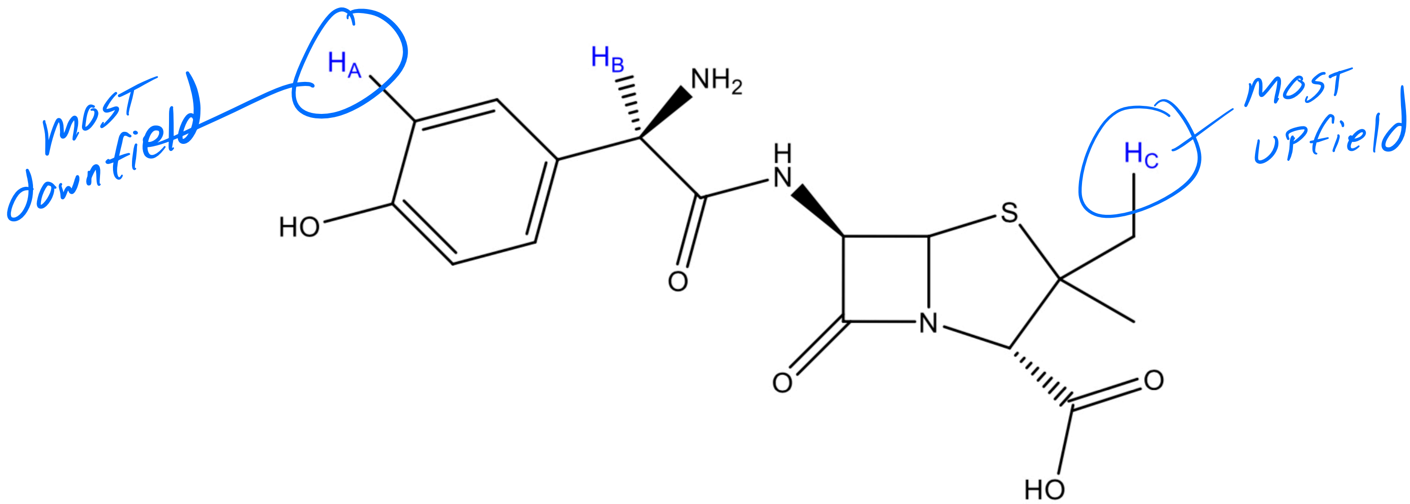




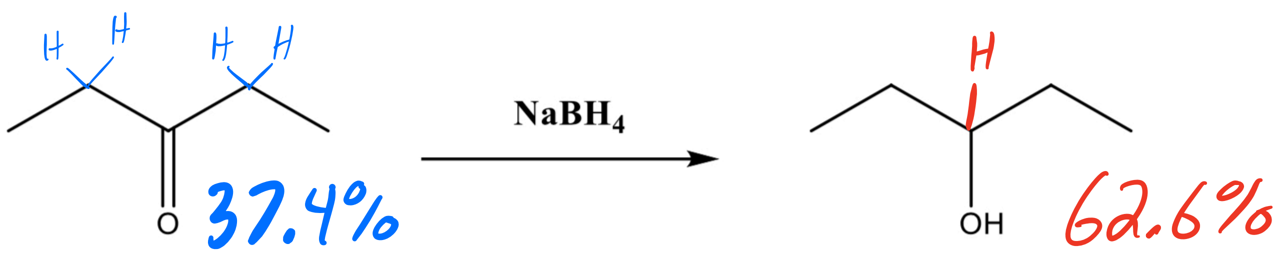
21. Assign the protons (labeled A-E) of the structure below to their corresponding signal on the NMR spectrum.

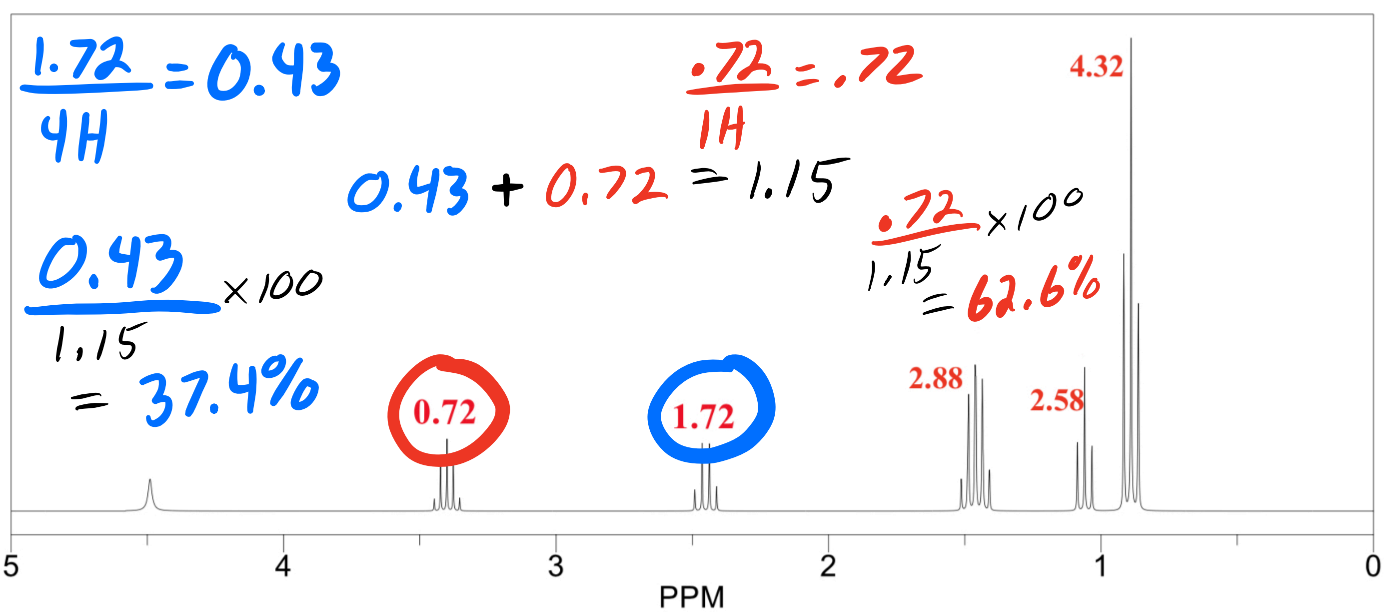


22. For each hydrogen atom (Ha, Hb, Hc) in the molecule below, rank the chemical shift from lowest (upfield) to highest (downfield).

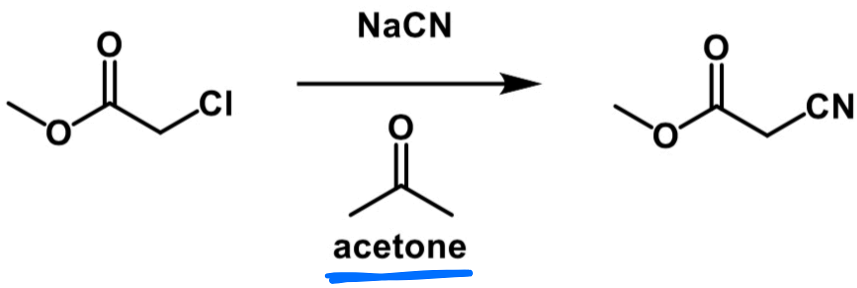


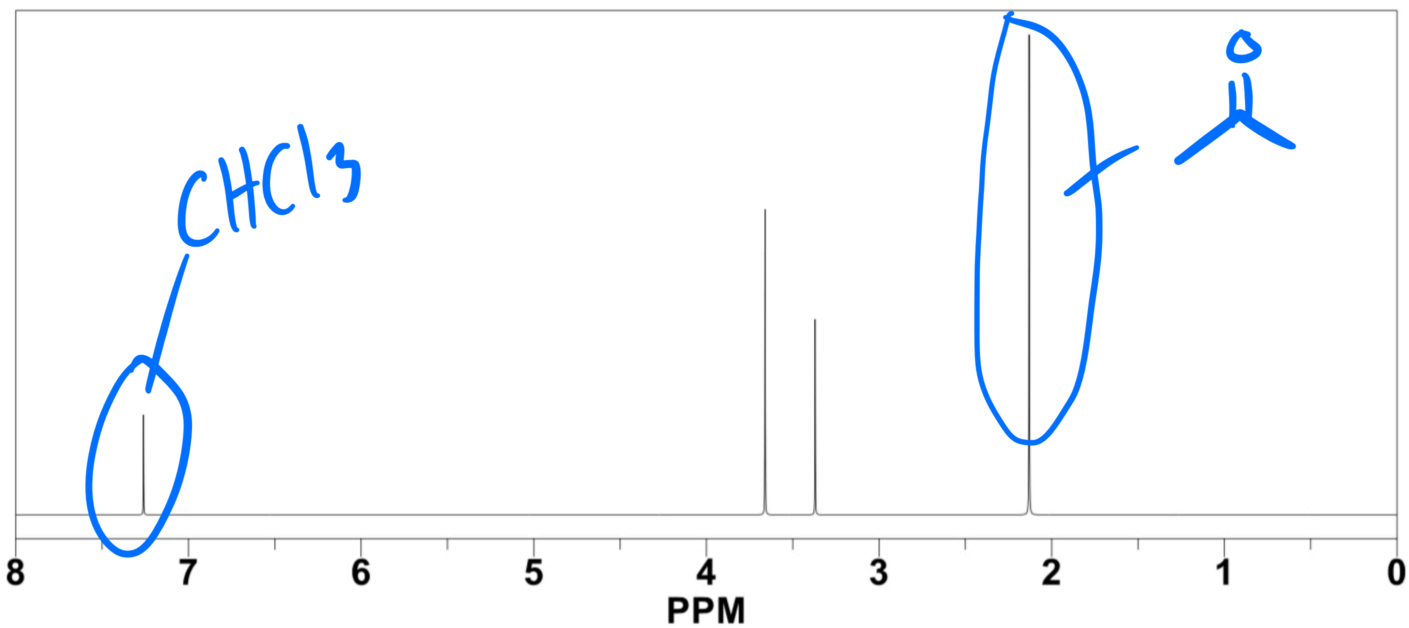
23. Consider the following reduction of 3-pentanone to form 3-pentanol. Calculate the percentage of each component of the mixture using the NMR spectrum.



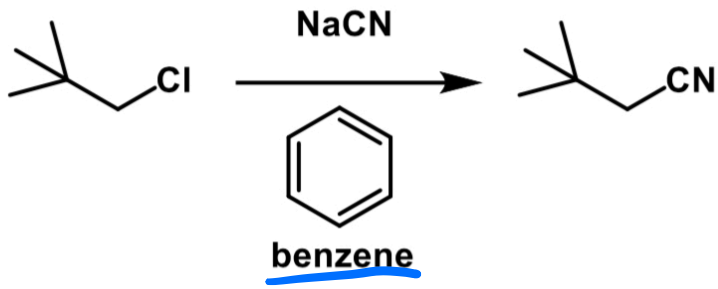


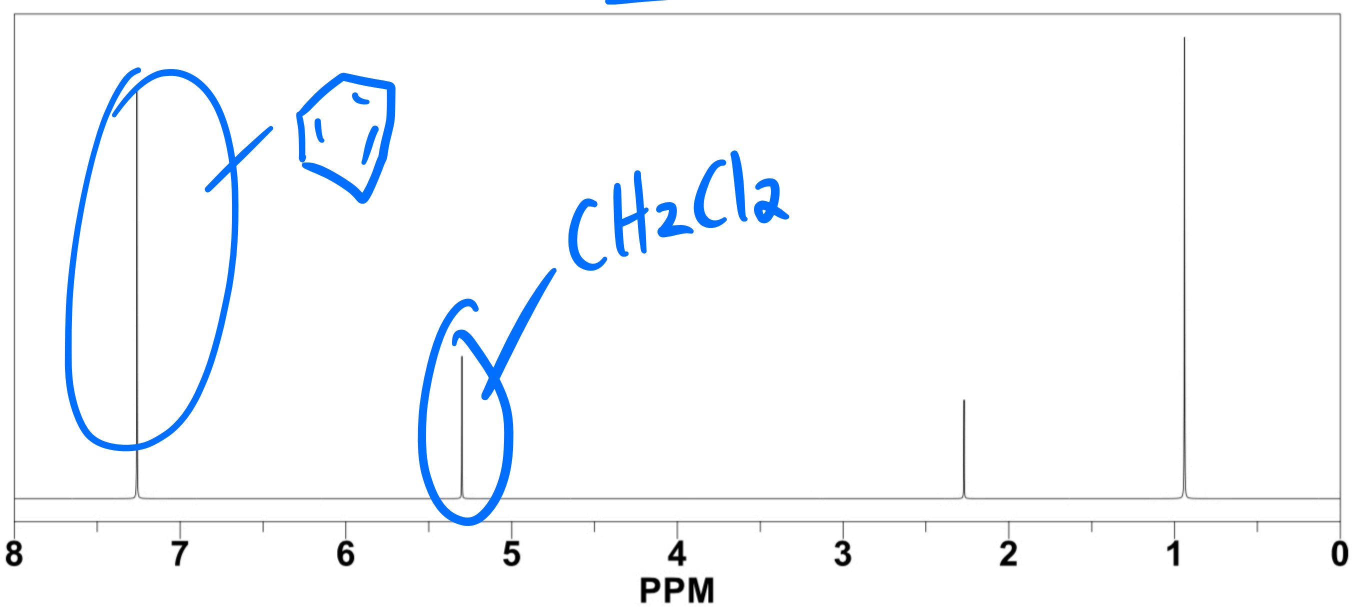
24. The following SN2 reaction was performed and monitored via TLC. After TLC verified that all of the starting material was consumed, the product was dissolved in deuterated chloroform, $CDCl_3$, and an NMR spectrum was obtained. Circle any residual solvent peaks present in the NMR spectrum.



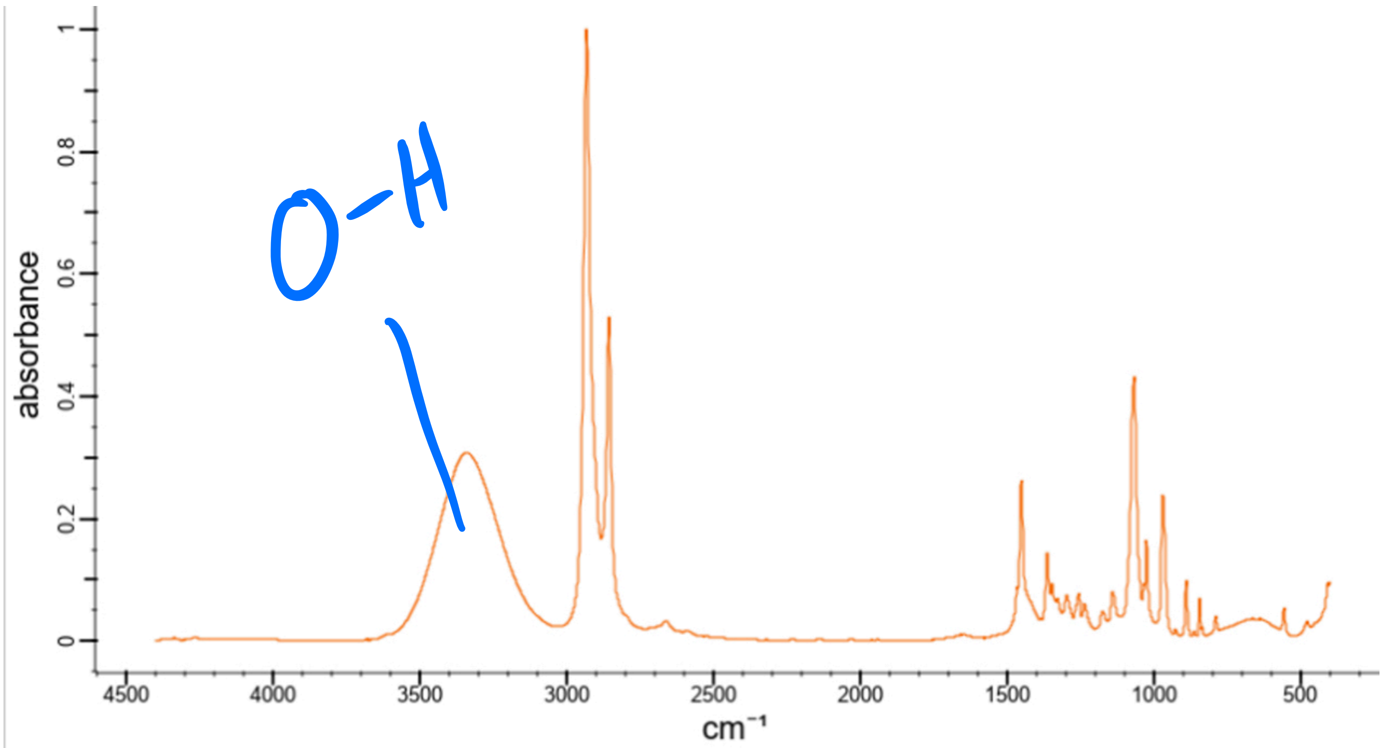


25. The following SN2 reaction was performed and monitored via TLC. After TLC verified that all of the starting material was consumed, the product was dissolved in deuterated dichloromethane, CD_2Cl_2, and an NMR spectrum was obtained. Circle any residual solvent peaks present in the NMR spectrum.





26. What type of spectroscopy is represented in the following spectrum? What does the signal at approximately 3300 cm^{-1} represent?



IR Spectroscopy